

# Clinical Validation of Sex-Specific Biomarker Thresholds in Cardiovascular Risk Assessment

The historical landscape of cardiovascular research has predominantly favored male-centric cohorts, leading to the establishment of diagnostic thresholds that often fail to account for the unique physiological and pathological trajectories of women. Current clinical practices frequently utilize "unisex" benchmarks for blood pressure, lipid profiles, and inflammatory markers, which may inadvertently lead to the underdiagnosis and undertreatment of female patients. Recent evidence from landmark trials, including the Systolic Blood Pressure Intervention Trial (SPRINT), the Women's Ischemia Syndrome Evaluation (WISE), and large-scale meta-analyses, suggests that cardiovascular risk manifests at lower absolute values in women across several key biomarkers. This report provides an exhaustive analysis of the evidentiary basis for sex-specific calibrations, evaluates the mechanisms of sexual dimorphism in vascular health, and recommends validated thresholds to bridge the existing diagnostic gap in female cardiovascular care.

## Sex-Specific Blood Pressure Thresholds and Vascular Aging

The paradigm of hypertension management is undergoing a significant shift as the medical community recognizes that the relationship between systolic blood pressure (SBP) and cardiovascular disease (CVD) risk is sexually dimorphic. While the current standard of  $\geq 130/80$  mmHg or  $\geq 140/90$  mmHg is used globally, observational and trial data indicate that women experience vascular damage and clinical events at pressures significantly lower than those typically identified as "high" in men.<sup>1</sup>

## Analysis of the SPRINT Trial and Sex-Specific Outcomes

The Systolic Blood Pressure Intervention Trial (SPRINT) was designed to evaluate whether a systolic target of  $< 120$  mmHg (intensive) was superior to  $< 140$  mmHg (standard) in reducing major cardiovascular events.<sup>3</sup> The overall trial findings demonstrated a 25% reduction in the primary composite endpoint and a 27% reduction in all-cause mortality.<sup>5</sup> However, the female representation in the trial was approximately 35.3%, leading to initial questions regarding the generalizability of these benefits to women.<sup>5</sup>

A critical subgroup analysis published by Foy et al. in *Circulation* (2018), titled "Are the Benefits of Intensive Systolic Blood Pressure Lowering Different in Women and Men?" examined these

sex-specific outcomes.<sup>5</sup> The primary composite endpoint in the intensive treatment group was reduced by **16%** in women compared to **27%** in men.<sup>5</sup> While the hazard ratio for women was **0.84 (95% CI, 0.61 to 1.13)**, it did not reach statistical significance, largely due to the lower number of events and the trial's early termination for benefit.<sup>5</sup> Despite this, the lack of statistical interaction between treatment effect and sex suggests that women likely derive a similar biological benefit from intensive lowering, even if the trial was underpowered to demonstrate it independently.<sup>5</sup>

Further post-hoc analysis by the SPRINT investigators revealed that women might be more sensitive to the duration of time spent in the target blood pressure range.<sup>6</sup> For every standard deviation increase in systolic blood pressure time in target range (SBP TTR), women experienced a statistically significant reduction in major adverse cardiovascular and cerebrovascular events (MACCE) with an adjusted hazard ratio of **0.85 (95% CI 0.74–0.99)**, whereas men only reached a similar trend when their TTR was significantly higher.<sup>6</sup> This implies that for women, maintaining a lower pressure consistently is even more prognostic for heart failure prevention than it is for men.<sup>8</sup>

## Quantifying Risk Equivalence and Lower Intervention Thresholds

Evidence from the Atherosclerosis Risk in Communities (ARIC) study and other multi-ethnic cohorts has provided a clearer picture of "risk equivalence" between sexes.<sup>1</sup> A pivotal study by Ji et al. (Circulation 2021) titled "Sex Differences in Blood Pressure Associations With Cardiovascular Disease" quantified the exact levels at which female risk matches male risk.<sup>1</sup>

The analysis found that an SBP of **100 to 109 mmHg** in women was associated with an incident CVD risk comparable to that of an SBP of **130 to 139 mmHg** in men.<sup>2</sup> This **20 to 30 mmHg** gap suggests that the current "Normal" range for men is actually a high-risk zone for women. For specific outcomes like myocardial infarction (MI), the risk for women with an SBP between **110 and 119 mmHg** was comparable to the risk for men with an SBP of **≥ 160 mmHg**.<sup>1</sup> Stroke risk manifested a similar pattern: women at **120–129 mmHg** had an equivalent risk to men at **140–149 mmHg**.<sup>1</sup>

These findings are corroborated by the Hordaland Health Study, which observed that Stage 1 hypertension (**130–139 mmHg**) was a significant predictor of acute coronary syndrome (ACS) in women (**HR 2.18, 95% CI 1.32–3.60**) but not in men (**HR 1.30, 95% CI 0.98–1.71**).<sup>9</sup> The biological rationale for this disparity is rooted in vascular anatomy; women possess smaller arterial diameters even after adjusting for body size, meaning that a given increase in pressure exerts a proportionally higher wall stress on the

female vasculature.<sup>1</sup>

### Validated Blood Pressure Thresholds for Female Diagnostics

Based on the evidence from Ji et al.<sup>1</sup> and the SPRINT subgroup analyses<sup>5</sup>, the current clinical practice of treating **128 mmHg** as "Normal" for women is likely delaying critical intervention. The research supports a downward shift of approximately **5 to 10 mmHg** across all zones to align with actual risk profiles.

| Zone       | Male Threshold (SBP) | Recommended Female Threshold (SBP) | Supporting Citations |
|------------|----------------------|------------------------------------|----------------------|
| Optimal    | <                    | <                                  | <sup>1</sup>         |
| Normal     | 120–130 mmHg         | 110–120 mmHg                       | <sup>1</sup>         |
| Borderline | 130–140 mmHg         | 120–128 mmHg                       | <sup>1</sup>         |
| High       | ≥                    | ≥                                  | <sup>2</sup>         |

### Lipid Profile Shifts and the Menopausal Transition

Lipid management in women is uniquely complicated by the menopausal transition, during which hormonal shifts drastically alter the lipid landscape. Estrogen is a key regulator of hepatic cholesterol metabolism, acting to increase the number of LDL receptors and facilitate the clearance of atherogenic particles.<sup>10</sup> The withdrawal of estrogen during menopause triggers a pro-atherogenic shift that often catches patients and clinicians off guard.<sup>10</sup>

### Quantifying the Post-Menopausal LDL Surge

Longitudinal studies, such as the Study of Women's Health Across the Nation (SWAN) and Chinese population cross-sectional data, have documented a significant rise in LDL cholesterol around the time of the final menstrual period (FMP).<sup>12</sup> LDL-C typically increases by approximately **10% to 15%** according to the *Archives of Medical Science* (2022), though some clinical reports indicate shifts as high as **15% to 25%**.<sup>10</sup> This increase often occurs rapidly, within the year before and after the FMP.<sup>12</sup>

For a woman entering menopause with an LDL of **100 mg/dL**, this surge can push her to

**120 or 125 mg/dL** within a matter of months, transitioning her from a "healthy" status to a "borderline" risk category.<sup>10</sup> Furthermore, menopause is associated with a shift in particle characteristics; LDL particles often become smaller and denser, which are more easily able to penetrate the arterial wall and initiate plaque formation.<sup>10</sup> Standard tests do not always capture this qualitative shift, leading to an underestimation of the newly acquired risk.<sup>10</sup>

## **Total Cholesterol Risk Curves and Diagnosis Disparities**

The Framingham Heart Study has long established that cholesterol-associated risk is not static across age or sex.<sup>14</sup> In the Framingham Risk Score, women are evaluated with a point system that accounts for the fact that a total cholesterol level of **200 to 239 mg/dL** in a 50-year-old woman carries different prognostic weight than it does in a man of the same age.<sup>14</sup> Data from the Copenhagen General Population Study shows that while the peak cholesterol rise occurs between ages 40 and 50 in men, it is delayed until ages 50 to 60 in women, closely mirroring the menopausal timeline.<sup>11</sup>

Evidence also suggests a diagnosis gap: women with confirmed cardiovascular disease often average higher cholesterol levels at the time of diagnosis than men at the same disease stage.<sup>11</sup> This is potentially because pre-menopausal women maintain lower baseline levels for decades, which creates a false sense of security that delays aggressive lipid-lowering therapy once post-menopausal levels escalate.<sup>11</sup>

## **Triglycerides as a Dominant Risk Factor in Women**

While LDL-C is the primary target of therapy, triglycerides (TG) appear to be a stronger independent risk factor for CVD in women than in men.<sup>13</sup> The prospective study of UK Biobank participants (2026) found that triglycerides were associated with larger hazard ratios for CVD in women than in men, with a women-to-men ratio of hazard ratios (RHR) of

**1.06 (95% CI, 1.02–1.09)**, indicating a **6%** excess risk for women per standard deviation increase.<sup>19</sup>

Mora et al. (Circulation 2008) highlighted that nonfasting triglycerides are a particularly potent predictor for women.<sup>18</sup> This may be due to sexual dimorphism in postprandial metabolism; men generally clear triglycerides faster, while women experience more prolonged postprandial elevations, peaking at 3–4 hours after a meal compared to 1–3 hours in men.<sup>20</sup> This extended exposure to triglyceride-rich lipoproteins (TRLs) and their remnants increases the atherogenic burden on the female endothelium.<sup>13</sup> Consequently, there is strong evidentiary support for a lower concern threshold for triglycerides in women, such as **175 mg/dL** for nonfasting samples, rather than the standard **200 mg/dL**.<sup>21</sup>

## **Corrected Lipid Thresholds for Female Risk Management**

Based on the rapid rise during menopause and the higher relative risk of triglycerides, the following thresholds are recommended to improve sensitivity for early detection of female CVD risk.

| Lipid Fraction                 | Male Threshold (mg/dL) | Recommended Female Threshold (mg/dL) | Supporting Citations |
|--------------------------------|------------------------|--------------------------------------|----------------------|
| Total Cholesterol (Borderline) | 200–240                | 190–220                              | 11                   |
| Total Cholesterol (High)       | $\geq$                 | $\geq$                               | 11                   |
| LDL (Borderline High)          | 130                    | 120                                  | 10                   |
| HDL (At Risk)                  | $<$                    | $<$                                  | 13                   |
| Triglycerides (Borderline)     | 150–200                | 150–170                              | 18                   |

## Diagnostic Accuracy of ST-Segment Depression in Women

The interpretation of the exercise stress test (EST) in women is notoriously difficult due to a high rate of false positives when using the standard male-calibrated **1.0 mm** ST-segment depression threshold.<sup>25</sup> This diagnostic challenge has historically contributed to "Yentl Syndrome," where women are misdiagnosed or receive less aggressive cardiac workups than their male counterparts.

### Findings from Gulati et al. (JAMA 2005)

The study by Gulati et al., "Exercise Capacity and the Risk of Death in Women," conducted as part of the St. James Women Take Heart Project, redefined the importance of stress test parameters.<sup>27</sup> The researchers found that in asymptomatic women, exercise capacity (measured in metabolic equivalents or METs) was a more powerful predictor of mortality than traditional ECG findings.<sup>27</sup> Specifically, for every 1-MET increase in exercise capacity, women had a **17%** reduction in the risk of death from any cause (**HR 0.83**).<sup>27</sup>

Regarding ST-segment depression, the study highlighted that a **1.0 mm** horizontal or downsloping depression has significantly lower specificity for obstructive coronary artery disease (CAD) in women.<sup>25</sup> While **1.0 mm** is the diagnostic "hallmark" in men, it often generates false positives in women due to factors like the "digoxin-like" effect of estrogen and the higher prevalence of resting ST-T wave abnormalities in the female population.<sup>25</sup> The mean sensitivity of the 1.0 mm threshold in women is approximately **61%** with a specificity of **70%**, both of which are markedly lower than the **68%** and **77%** observed in men.<sup>26</sup>

## Insights from the WISE Study

The Women's Ischemia Syndrome Evaluation (WISE) study further clarified these diagnostic discrepancies.<sup>29</sup> WISE demonstrated that many symptomatic women who exhibit ST depression on a stress test do not have significant epicardial stenosis on angiography, but instead suffer from coronary microvascular dysfunction (CMD).<sup>29</sup> Because standard ECG thresholds are designed to detect epicardial obstruction, they often miss the microvascular disease more common in women or erroneously flag non-ischemic repolarization changes.<sup>29</sup>

To improve specificity for obstructive CAD in female patients, many experts suggest a higher threshold of **1.5 mm** for concern.<sup>26</sup> This calibration reduces the false positive rate, ensuring that women who are referred for invasive angiography are truly at high risk for major coronary blockages.<sup>26</sup>

| Parameter                   | Standard (Male) | Corrected (Female) | Source Citation |
|-----------------------------|-----------------|--------------------|-----------------|
| ST Depression (Normal)      | <               | <                  | 25              |
| ST Depression (Moderate)    | 1.0–2.0 mm      | 1.5–2.5 mm         | 25              |
| ST Depression (Significant) | ≥               | ≥                  | 25              |
| Sensitivity (1.0 mm)        | 68%             | 61%                | 26              |
| Specificity (1.0 mm)        | 77%             | 70%                | 26              |

# HbA1c and the "Diabetic Risk Multiplier" in Women

While the biochemical thresholds for diagnosing pre-diabetes and diabetes are standardized ( 5.7% and 6.5% ), the relative impact of hyperglycemia on the cardiovascular system is far more severe in women.<sup>31</sup>

## Analysis of Peters et al. (Diabetologia 2014)

A landmark systematic review and meta-analysis by Peters et al. (2014) involving over 850,000 individuals found that type 2 diabetes confers a significantly higher excess risk of coronary heart disease in women than in men.<sup>32</sup> The presence of diabetes nearly tripled the risk of incident CHD in women (RR 2.82, 95% CI 2.35, 3.38 ), while it only doubled the risk in men (RR 2.16, 95% CI 1.82, 2.56 ).<sup>32</sup> This indicates a 44% greater relative excess risk for female diabetics.<sup>32</sup>

The disparity in stroke risk is also substantial. Diabetic women have a 27% higher risk of stroke compared with their male counterparts.<sup>31</sup> These findings suggest that the biological "female advantage" regarding heart disease—partially conferred by estrogen before menopause—is completely erased once a woman develops diabetes.<sup>31</sup>

## Pre-Diabetes and Early Risk Manifestation

The cardiovascular damage associated with glucose dysregulation begins well before the threshold for overt diabetes is met. Research suggests that during the transition from normoglycemia to impaired glucose tolerance (pre-diabetes), women experience a sharper decline in their metabolic and vascular risk factor profile than men.<sup>31</sup> Women in the pre-diabetic range ( 5.7%–6.4% ) show higher rates of endothelial dysfunction and systemic inflammation than age-matched men in the same range.<sup>31</sup>

Consequently, some investigators recommend a risk multiplier in clinical risk scoring tools for women with HbA1c  $\geq 5.7\%$  .<sup>35</sup> While men might require a  $2x$  multiplier to their baseline risk if diabetic, women may require a multiplier of  $4x$  to  $5x$  to accurately reflect their clinical outcomes.<sup>35</sup>

| Metric | Relative Risk (Men) | Relative Risk (Women) | Excess Risk in Women |
|--------|---------------------|-----------------------|----------------------|
|        |                     |                       |                      |

|              |      |      |            |
|--------------|------|------|------------|
| Incident CHD | 2.16 | 2.82 | 44% higher |
| Fatal Stroke | 1.83 | 2.28 | 27% higher |

## Inflammatory Markers: hs-CRP and Sex-Specific Hazards

High-sensitivity C-reactive protein (hs-CRP) has gained prominence as a non-lipid marker for subclinical inflammation and cardiovascular risk.<sup>36</sup>

### JUPITER Trial Data and hazard Ratios

The JUPITER trial demonstrated that in individuals with "normal" LDL ( $< 130 \text{ mg/dL}$ ), an elevated hs-CRP ( $\geq 2.0 \text{ mg/L}$ ) was a significant marker for future cardiovascular events that could be mitigated by statin therapy.<sup>38</sup> While Ridker et al. did not explicitly set different cutoffs by sex in the primary trial design, subsequent analyses revealed that hs-CRP may be more predictive of cardiac events in women.<sup>36</sup>

Hazard ratios for a 1-standard deviation increase in hs-CRP were found to be approximately 1.4, a magnitude that is often larger than that associated with a similar increase in cholesterol.<sup>40</sup> In the Women's Health Study, women with low LDL but high hs-CRP had higher absolute risk than those with high LDL but low hs-CRP, highlighting the potency of this inflammatory biomarker in female populations.<sup>36</sup>

### Guideline Recommendations for hs-CRP

The 2026 ACC/AHA lipid guidelines identify hs-CRP  $\geq 2.0 \text{ mg/L}$  as a critical "risk enhancer".<sup>41</sup> In patients at "borderline" 10-year risk (3%–5%), an hs-CRP  $\geq 2.0 \text{ mg/L}$  measured on two successive occasions justifies the initiation of high-intensity statin therapy.<sup>41</sup> For women, who often have lower absolute 10-year risk scores due to their age, this "borderline" category is common, making the hs-CRP threshold of  $2.0 \text{ mg/L}$  a vital tool for preventing under-treatment.

## Anthropometry: Waist Circumference vs. BMI

Body mass index (BMI) is increasingly criticized as a cardiovascular predictor because it does



not account for sex-specific fat distribution.<sup>42</sup> In women, central adiposity (visceral fat) is a far better predictor of CVD than total body weight.<sup>43</sup>

## IDF and WHO Source Thresholds

The International Diabetes Federation (IDF) and the World Health Organization (WHO) have formalized sex-specific thresholds for waist circumference (WC) as a mandatory component of metabolic syndrome diagnosis.<sup>42</sup> For populations of European origin, the IDF recommends a threshold of  $\geq 94 \text{ cm}$  for men and  $\geq 80 \text{ cm}$  for women.<sup>42</sup>

These thresholds highlight the massive disparity in risk: an  $88 \text{ cm}$  waist measurement is considered "normal" or "low risk" for men in many US clinical settings but is classified as "high risk" for women.<sup>42</sup> Recent data from sub-Saharan African and Asian populations confirm that a waist-to-height ratio (WHtR) of  $\geq 0.5$  or a sex-specific WC threshold is more sensitive than BMI for identifying hypertension and dyslipidemia in female patients.<sup>44</sup>

## Risk-Equivalent BMI

Research on "risk-equivalent" BMI thresholds suggests that women may reach a high risk of cardiovascular disease and diabetes at lower BMI values than men, especially in non-white populations.<sup>47</sup> In South Asian and East Asian women, a risk-equivalent BMI (comparable to  $30 \text{ kg/m}^2$  in White women) was found to be as low as  $21 \text{ kg/m}^2$ .<sup>47</sup> This underscores the need for a "BMI Reliability Warning" in diagnostic apps, noting that central adiposity measurements should always supersede BMI in female risk assessment.<sup>42</sup>

## Conclusion: Bridging the Female Diagnostic Gap

The evolution of cardiovascular diagnostics necessitates a departure from unisex benchmarks.

The evidence analyzed in this report—from the SBP risk-equivalence at  $100 \text{ mmHg}$  for women to the  $44\%$  excess risk associated with diabetic hyperglycemia—demonstrates that female physiology requires a more sensitive diagnostic net.<sup>1</sup> By adopting the sex-specific thresholds for blood pressure, lipid profiles, and ST-segment interpretation detailed herein, clinical tools can improve the early detection of subclinical disease and potentially mitigate the historical trend of higher post-event mortality in female cardiac patients.<sup>5</sup> The integration of these validated markers, supported by the 2026 guidelines and decades of observational data, represents a critical step toward precision medicine in women's heart health.

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